

Media Engineering and Technology Faculty
German University in Cairo



Algebraic Logic of Secrets

Bachelor Thesis

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Supervisors: Prof. Haythem O. Ismail

Submission Date: 29 May, 2025

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This is to certify that:

- (i) the thesis comprises only my original work toward the Bachelor degree
- (ii) due acknowledgment has been made in the text to all other material used

Omar Gamal Eldin
29 May, 2025

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Abstract

As artificial agents become increasingly integrated into our daily lives, it is essential that they are capable of understanding and reasoning about complex social phenomena. Secrecy is one of the most fundamental of these, playing a key role in trust, communication, and decision-making. This thesis introduces the *Algebraic Logic of Secrets*, a formal framework for modeling secrecy over time in multi-agent systems.

Building on modal and algebraic logic, the framework synthesizes components from the Commonsense Theory of Secrets, Versatile Event Logic (VEL) for temporal reasoning, and algebraic treatments of belief and state. Secrecy is formalized as a structured relation involving propositions, agents, group membership, time, belief, intention, and revelation. By avoiding possible-world semantics and adopting a compositional algebraic approach, the logic supports a more realistic and explainable reasoning model.

The framework includes key axioms and theorems, and examines the effect of removing logical omniscience to better capture the behavior of resource-limited agents. This work provides a formal foundation for enabling artificial agents to reason about secrets in a structured and socially-aware manner.

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Chapter 1

Introduction

As artificial agents become more embedded in human environments, their ability to understand and reason about social concepts is no longer optional. In particular, secrecy plays a critical role in trust, privacy, and communication, making it essential for agents to grasp and handle secret information appropriately.

Although logic-based approaches to agent reasoning have matured over the years, secrecy has not received the same attention, especially from a commonsense or multi-agent perspective. Existing work often focuses on narrow contexts like cryptographic protocols, leaving a gap in modeling how secrets function in everyday human-agent or agent-agent interactions. This thesis aims to help bridge that gap.

1.1 Motivation

In settings where trust, privacy, and communication are essential, agents must be able to recognize, respect, and act upon the presence of secrets. Whether assisting users, collaborating with other agents, or operating in sensitive domains, understanding secrecy is central to behaving appropriately in real-world situations.

Despite its relevance, secrecy remains an underexplored concept in formal logic. Existing treatments are often limited to security protocols or rely on idealized assumptions about agent knowledge. These models fall short when applied to agents interacting with humans in dynamic, uncertain environments.

This thesis aims to address that gap by developing a logic that captures how secrecy operates over time, how agents reason about what is known and unknown, and how belief, intention, and revelation interact in structured secrecy situations. A formal, algebraic treatment of these concepts is necessary not only for theoretical clarity but also for practical applications in explainable AI and secure, socially intelligent systems.

1.2 Aim

The aim of this thesis is to develop a formal, algebraic logic for representing and reasoning about secrets in multi-agent systems. Specifically, we seek to construct a framework that integrates belief, intention, revelation, and time within a unified structure, without relying on possible-world semantics.

By extending existing algebraic logics of belief and state, and incorporating the Commonsense Theory of Secrets and Versatile Event Logic, this work introduces the *Algebraic Logic of Secrets* (Log_ASec). The goal is to provide a compositional and expressive system capable of modeling how secrecy is maintained, violated, or reasoned about by artificial agents over time.

This logic is intended to support both theoretical investigation and future practical implementations in areas such as explainable AI, secure agent interaction, and cognitive modeling.

1.3 Thesis Outline

The chapters of this thesis are organized as follows:

- **Chapter 1: Introduction**
Introduces the topic, motivation, aim, and overall structure of the thesis.
- **Chapter 2: The Theory of Secrets**
Presents the Commonsense Theory of Secrets and defines secrets using belief, intention, and revelation. Introduces Versatile Event Logic (VEL) to model secrecy over time.
- **Chapter 3: Algebraic Logic**
Develops the Algebraic Logic of Belief (Log_AB) and the Algebraic Logic of States (Log_AS), providing the formal foundations for reasoning without possible-world semantics.
- **Chapter 4: Algebraic Logic of Secrets**
Defines the Log_ASec framework, including its language, semantics, and axioms. Demonstrates how it models secrecy and analyzes the impact of removing logical omniscience through selected theorem proofs.
- **Chapter 5: Conclusion and Future Work**
Summarizes the contributions of the thesis and suggests directions for implementation and further research.
- **Appendix A: Propositional and Modal Logic**
Reviews propositional and modal logic to support the formal background of the thesis.

Chapter 2

The Theory of Secrets

Secrets play a fundamental role in social and epistemic interactions. They can range from personal information, such as passwords and private confessions, to state and corporate secrets that influence global affairs. Understanding the nature of secrets is crucial for fields such as artificial intelligence, information security, and social reasoning. This chapter presents an extended formalization of secrecy based on *A Commonsense Theory of Secrets* [11].

2.1 Intuitions About Secrets

Before formalizing the logic of secrets, it is useful to outline key commonsense intuitions that underpin secrecy. These intuitions help distinguish between different aspects of secrets and their logical structure.

Definition (Secretum and Secreta). Objects of secrecy, termed *secretum* (plural of *secretum*), are the propositions representing confidential information. Secrets are not these objects themselves, but relations between *secretum* and other entities, such as secret keepers and nescients.

- **Secrets are distinct from secretum.** A secret is not the object itself (e.g., a document or password), but rather a relation between the *secretum* and other entities.
- **Secretum are propositions.** The content of a secret is always a proposition, not a physical entity. For example, a classified military map is not itself a secret, but the fact that it represents a certain location is.
- **Secrets require secret keepers.** A piece of information is only a secret if at least one agent is actively keeping it confidential.
- **Secrets require nescients.** There must be at least one agent (a “nescient”) from whom the secret is intentionally concealed.

- **Secret keepers and nescients form structured groups.** These groups may be fixed (extensional) or dynamic (intensional), with membership changing over time.
- **Secrets exist only under a secrecy condition.** A proposition remains a secret only as long as a certain condition holds (e.g., security clearance, a time limit, or an agreement among keepers).
- **Secrets are temporary.** No secret lasts forever; they either expire naturally or are eventually revealed.
- **Secret keepers believe the secret and its secrecy condition.** Every secret keeper must believe the proposition they are keeping secret and the condition under which it remains a secret.
- **Secret keepers do not believe the secret has been revealed to a nescient.** If a secret keeper believes the information has already been revealed, they no longer maintain it as a secret.
- **Secret keepers intend to keep the secret.** A defining characteristic of secrecy is the intention of the secret keepers to prevent the nescients from learning the secret while the secrecy condition holds.

These intuitions serve as the foundation for the logical formalization of secrets presented in the next section.

2.2 Formal Representation of Secrets

Before defining secrecy in logic, we need to describe what a secret consists of. This section outlines the main elements involved in representing a secret formally.

2.2.1 Structure and Formalization of Secrets

A secret is not merely an unknown fact; rather, it is a structured relation involving several key components [11]. These components provide the foundation for modeling secrecy both conceptually and formally:

- **Secretum (Object of the Secret):** The proposition ϕ that constitutes the secret. Secreta are always propositions rather than physical entities. This distinction ensures that secrecy is not about the physical concealment of objects but about restricting access to information.
- **Secret Keepers (K):** A group of agents who are aware of ϕ and are responsible for keeping it secret. These agents actively intend to prevent the disclosure of ϕ to unauthorized individuals.

- **Nescients (N):** A group of agents who do not know ϕ and from whom the information is intentionally concealed. Nescients may actively seek to uncover ϕ , or they may be passively unaware.
- **Secrecy Condition (ψ):** A condition that governs the persistence of secrecy. If ψ ceases to hold, the secret may no longer be considered as such. This may represent time constraints, legal agreements, or contextual conditions.
- **Time (t):** The moment at which secrecy is evaluated. Secrets are dynamic and may change depending on time and agent knowledge.

We write $\text{Secret}(\phi, K, N, C, t)$ to state that, at time t , proposition ϕ is kept a secret by group K from group N while the secrecy condition C holds. This notation reflects the core structure of a secrecy situation without committing to a specific formalization. The full logical characterization of secrecy may impose additional constraints on belief, intention, and revelation, but we reserve such formal treatment for later sections.

2.2.2 Logical Representation of Secrets

To formally define secrecy, we employ a **First-Order Modal Logic (FOML)** framework that extends **Propositional Modal Logic (PML)** with quantification over agents and temporal dynamics [4]. The logical syntax used to describe secrecy is:

$$\phi := P \mid \neg\phi \mid \phi \wedge \phi \mid \forall x[\phi] \mid \boxed{*}\phi \mid H(\phi, t) \mid B(A, \phi) \mid I(A, \phi) \mid R(A, \phi)$$

where:

- P represents atomic propositions.
- $\neg\phi$, $\phi \wedge \phi$, and $\forall x[\phi]$ follow standard logical operations.
- $\boxed{*}\phi$ represents necessity across histories that share the same past.
- $H(\phi, t)$ states that ϕ **holds at** time t .
- $B(A, \phi)$ means agent A **believes** ϕ .
- $I(A, \phi)$ means agent A **intends** ϕ .
- $R(A, \phi)$ means ϕ has been **revealed** to agent A . Note that revelation does not necessarily imply that agent A believes ϕ [11].

Additionally, we introduce the membership predicate $Mem(A, G)$, indicating that agent A is a **member** of group G .

To simplify notation, we use the following shorthand: If O is B , I , R , or Mem , where α and β are elements such as agents, groups, propositions, and t is a time parameter of type σ_T , then we write $O(\alpha, \beta, t)$ as a shorthand for $H(O(\alpha, \beta), t)$. This provides a more concise representation of time-dependent beliefs, intentions, revelations, and group membership [11].

The **modal properties of belief and intention** follow the axioms discussed in **Appendix A**, where belief is modeled as a **KD45 system**, ensuring consistency and introspection, while intention follows a **KD structure**. This logical foundation ensures that secret keepers' reasoning is formally structured [13].

2.2.3 Types of Secrets

Different types of secrecy arise from variations in the conditions of belief, intention, and revelation [11].

Secret₀ represents the most fundamental form of secrecy, ensuring that secret keepers believe a proposition ϕ , intend to keep it secret, and do not believe it has been revealed to nescients. However, most secrets involve additional conditions that strengthen this basic notion. Five stronger types of secrecy, all implying **Secret₀**, are defined below [11, 7]:

Secret₁: The secretum ϕ is not actually revealed to the nescients, meaning there is no instance where a nescient has learned the secret.

Secret₂: Every secret keeper is aware that they are keeping a secret, ensuring that the act of secrecy is consciously recognized among all members of the keeper group.

Secret₃: Secret keepers believe that the secretum ϕ has not been revealed to the nescients. This means that not only is secrecy being maintained, but the keepers also have confidence that it has not been compromised.

Secret₄: Secret keepers are aware of the identity of all other secret keepers, ensuring a shared understanding of who is involved in maintaining the secret.

Secret₅: Secret keepers believe that all other keepers are aware of the secrecy, meaning they not only recognize their own role in secrecy but also trust that other keepers acknowledge the same commitment.

Henceforth, we denote these types as S_1, S_2, S_3, S_4 , and S_5 respectively. In many real-world secrecy scenarios, secrets satisfy multiple or all of these conditions, meaning they fulfill the criteria for several of these secrecy types simultaneously.

These secrecy types are interrelated [11]:

S_2 and S_3 hold if and only if a secret exists and every secret keeper believes the secret and that it has not been revealed (S_1). S_2 follows directly from S_5 , since if all keepers believe the secrecy, they must also be aware that they are keeping it. S_4 implies that S_5 holds if and only if all keepers believe both the secret and that all co-keepers believe it (S_2). If the secrecy condition explicitly states that the secret is not revealed to nescients, then S is equivalent to S_3 . These distinctions clarify various levels of secrecy and how they function within social and epistemological contexts.

2.3 Modeling Time in Secrets using VEL

The logic of secrets requires a structured representation of time to model how secrecy persists or changes over time. To achieve this, we employ the Versatile Event Logic (VEL) formalism, introduced in [1]. VEL provides a branching-time framework where each moment in time has multiple possible future states, allowing us to represent how secrets evolve depending on different actions and conditions. The branching structure of VEL allows us to represent how secrets persist, become obsolete, or are revealed over time. In this framework, a secret is not merely a static piece of information but an event-dependent condition. This highlights the advantage of using VEL, as it enables the specification of how knowledge conditions evolve dynamically across different timelines.

2.3.1 Formal Representation of Events and States in VEL

VEL models time as a tree of possible histories, where:

- Each state has a unique past but multiple possible futures [1].
- Events occur at specific points in time and influence the system's state [1].
- Temporal persistence ensures that a fluent remains true unless explicitly terminated by an event, following the inertia principle from Event Calculus [1, 12].

VEL represents events as transitions between states, meaning that a secret remains unchanged unless an event explicitly alters it. This allows for:

- Modeling secrecy over time, where a secret persists unless an event (e.g., a leak or deliberate revelation) modifies it.
- Defining secrecy violations using the “Occurs” relation to capture when a revealing event takes place:

$$Occurs(e, \delta)$$

which states that an event e occurs over the interval δ , beginning at $b(\delta)$ and ending at $e(\delta)$ [1].

- Analyzing alternative futures where different event outcomes determine whether secrecy is maintained or lost [1].

2.3.2 Interpretation of VEL in Time and Events

VEL introduces the **Holds-at** predicate to express whether a proposition remains true at a given time:

$$H(\phi, t)$$

where $H(\phi, t)$ states that ϕ holds at time t in the actual history [1].

VEL defines a **History Structure** as a tuple:

$$\mathcal{H} = \langle S, T, \prec, H \rangle$$

where:

- S is the set of world states.
- T is the set of time points, shared across all histories.
- $\prec$ is a linear (irreflexive) order on T , defining temporal succession.
- H is the set of histories, where each $h \in H$ is a function $h : T \rightarrow S$, mapping each time point to a world state.

Each **index point** $\langle h, t \rangle$ represents a unique moment in a particular history, i.e., the state of the world $h(t)$ at time t along history h . This distinction is essential in branching-time semantics: while time points $t \in T$ are shared across all histories, the states at those times can differ depending on the path taken through the tree of possibilities.

By integrating VEL into our secrecy framework, we formally describe when a secret remains valid, when it is revealed, and how its status changes based on agent actions. This structured approach provides a logical foundation for analyzing secrecy over time, incorporating both persistence and change.

2.4 Conclusion

This chapter has formalized the commonsense theory of secrets using a structured modal logic framework. By distinguishing between belief, intention, and revelation, we capture the dynamic nature of secrecy in social reasoning. This framework allows for a precise understanding of how secrets are maintained, revealed, and structured across different agents and conditions. The next chapters will explore algebraic logic approaches to secrecy, comparing them to the modal logic framework developed here.

Chapter 3

Algebraic Logic

Algebraic logic provides a formal framework for reasoning about secrecy by structuring logical operations as algebraic transformations rather than relying on relational models. This approach is particularly suited for analyzing knowledge, belief, and state dynamics in secrecy modeling. Unlike modal logic, which represents secrecy through possible-world semantics, algebraic logic offers a more compositional method, capturing secrecy conditions through algebraic structures that explicitly define operations on knowledge [6]. Algebraic logic offers an alternative approach to secrecy analysis, addressing some limitations of modal logic in handling complex secrecy conditions and belief interactions, though it is not strictly necessary.

3.1 Modal and Algebraic Logic Comparison

Modal logic and algebraic logic are two distinct approaches to formal reasoning, each offering unique advantages and limitations when applied to secrecy modeling. Modal logic is traditionally used to express necessity and possibility through relational semantics, whereas algebraic logic provides a structural, compositional framework for reasoning about logical properties.

One of the challenges in secrecy modeling is capturing how information propagates while preserving constraints on access. Modal logic provides a way to express constraints on information flow using necessity and possibility operators, but secrecy requires additional assumptions about the underlying accessibility relations [5].

While modal logic remains a powerful tool for formalizing secrecy dynamics, algebraic logic offers a complementary approach with significant advantages in secrecy modeling. By eliminating the need for explicit possible-world semantics and leveraging algebraic structures, secrecy conditions can be more efficiently represented and manipulated.

To develop an algebraic framework for secrecy, we first establish the foundations of algebraic logic for belief and states. The next sections will introduce the **Algebraic**

Logic of Belief ($Log_A B$) and the **Algebraic Logic of States** ($Log_A S$), which provide the necessary formal tools for reasoning about epistemic and temporal aspects of secrecy [9, 10]. These components will then be unified to construct the **Algebraic Logic of Secrets**, a framework for modeling secrecy in an algebraic setting.

3.2 Syntax and Semantics of Algebraic Logic

The syntax and semantics of Algebraic Logic establish a common foundation used by both **Algebraic Logic of Belief** ($Log_A B$) and **Algebraic Logic of States** ($Log_A S$). Both logics share a many-sorted first-order algebraic structure where expressions are terms rather than sentences.

3.2.1 Syntax

Algebraic Logic syntax comprises the following core components:

- **Sorts:** Each logic partitions terms into base syntactic sorts:
 - Individual terms (σ_I)
 - Function types ($\tau_1 \rightarrow \tau_2$), where $\tau_1, \tau_2 \in \sigma$
- **Alphabet:** Defined as a union of four disjoint sets:
 - **Constants and function symbols** (Ω): A non-empty, countable set of symbols with designated syntactic types.
 - **Variables** (Ξ): Countably infinite set of variables partitioned according to sorts.
 - **Syncategorematic symbols** (Σ): Including parentheses, commas, and the universal quantifier ($\forall$).
 - **Logical symbols** (Λ): Boolean algebra operators (negation $\neg$, conjunction $\wedge$, disjunction $\vee$).
- **Formation rules:** Terms in algebraic logic are recursively formed by:
 - Variables and constants.
 - Function applications based on type assignments.
 - Boolean combinations using logical symbols.
 - Quantification ($\forall$).

3.2.2 Semantics

Semantically, Algebraic Logic utilizes structures built on Boolean algebras [6]. An algebraic structure $\mathfrak{S}$ is defined generally as:

- $\mathcal{D}$: Domain partitioned into subsets specific to each logic.
- $\mathfrak{A}$: Complete Boolean algebra $\langle P, +, \cdot, -, \perp, \top \rangle$, where:
 - P : the set of propositions.
 - $+$: disjunction (join).
 - $\cdot$: conjunction (meet).
 - $-$: negation.
 - $\perp$: the least element (false).
 - $\top$: the greatest element (true).

Valuation provides a variable assignment and an interpretation for constants and function symbols. Specifically:

- A variable assignment v maps each variable to a domain element.
- Each constant symbol is mapped to a fixed domain element.
- Each function symbol is interpreted as a domain function of appropriate arity.

The interpretation of terms is defined recursively:

- $[[x]]^v = v(x)$ for variables x
- $[[c]]^v$ is the domain element assigned to constant c
- $[[f(t_1, \dots, t_n)]]^v = f^I([[t_1]]^v, \dots, [[t_n]]^v)$
- Boolean operators follow standard algebraic semantics:

$$[[\neg t]]^v = -[[t]]^v, \quad [[t_1 \wedge t_2]]^v = [[t_1]]^v \cdot [[t_2]]^v, \quad [[t_1 \vee t_2]]^v = [[t_1]]^v + [[t_2]]^v.$$

- Universal quantification is interpreted as a meet over all domain assignments:

$$[[\forall x(t)]]^v = \prod_{a \in D_{\sigma(x)}} [[t]]^{v[a/x]}$$

Logical Consequence Logical consequence in Algebraic Logic is defined algebraically:

$$\Gamma \models \phi \quad \text{iff} \quad \prod_{\gamma \in \Gamma} [[\gamma]]^V \leq [[\phi]]^V$$

The symbol $\leq$ denotes the partial order of the Boolean algebra, meaning that the conjunction of all premises semantically entails the conclusion.

Properties of Logical Consequence:

1. If $\phi \in \Gamma$, then $\Gamma \models \phi$.
2. If $\Gamma \models \phi$ and $\Gamma \subseteq \Delta$, then $\Delta \models \phi$.
3. If $\Gamma \models \psi$ and $\Gamma \cup \{\psi\} \models \phi$, then $\Gamma \models \phi$.

3.3 Algebraic Logic of Belief

Traditional approaches to the logic of belief fall into two categories: **modal logic** and **first-order syntactical logic**. Modal logic models belief using possible-world semantics, while syntactical first-order logic employs self-referential sentences. However, both approaches have limitations: modal logic lacks expressiveness, and first-order syntactical logic suffers from paradoxes like the Liar Paradox [3, 4].

Unlike traditional belief logics, which rely on modal accessibility relations or syntactic self-reference, the **Algebraic Logic of Belief** ($Log_A B$) provides a purely algebraic alternative. By treating belief as a function in a Boolean algebra rather than a syntactic or modal operator, $Log_A B$ avoids certain paradoxes found in modal and syntactical approaches, such as self-referential paradoxes, while maintaining first-order expressiveness [9].

3.3.1 An Algebraic Approach to Belief

$Log_A B$ introduces a non-paradoxical, algebraic framework for reasoning about belief, where all expressions are terms that denote propositions rather than formulas. Its fundamental structure is based on three key principles. First, propositions in $Log_A B$ form a Boolean algebra, ensuring that logical operations are well-defined and free from paradoxes. Second, belief is represented as a function $B(a, \phi)$, where a is an agent and ϕ is a proposition, rather than being modeled through modal operators or predicates. Finally, since $Log_A B$ does not treat propositions as syntactical objects, it avoids self-referential paradoxes, providing a more consistent foundation for reasoning about belief [9].

Property	$Log_A B$ Term
Meet-distributivity	$\forall a, p_1, p_2 [(B(a, p_1) \wedge B(a, p_2)) \Leftrightarrow B(a, p_1 \wedge p_2)]$
Join-distributivity	$\forall a, p_1, p_2 [(B(a, p_1) \vee B(a, p_2)) \Leftrightarrow B(a, p_1 \vee p_2)]$
Conceit	$\forall a, p [B(a, p) \Rightarrow p]$
Consistency	$\forall a, p [B(a, \neg p) \Rightarrow \neg B(a, p)]$
Autoepistemology	$\forall a, p [\neg B(a, p) \Rightarrow B(a, \neg p)]$
Positive introspection	$\forall a, p [B(a, p) \Rightarrow B(a, B(a, p))]$
Negative introspection	$\forall a, p [\neg B(a, p) \Rightarrow B(a, \neg B(a, p))]$
Positive faithfulness	$\forall a, p [B(a, B(a, p)) \Rightarrow B(a, p)]$
Negative faithfulness	$\forall a, p [B(a, \neg B(a, p)) \Rightarrow \neg B(a, p)]$

3.3.2 Properties of Belief in $Log_A B$

$Log_A B$ provides an algebraic characterization of common belief properties, avoiding issues like logical omniscience and paradoxes [9]. The key belief properties formalized in [9] include :

3.3.3 Syntax and Semantics of $Log_A B$

Syntax:

$Log_A B$ is a many-sorted first-order algebraic logic in which all expressions are terms rather than formulas [9]. Its core syntactic sorts are:

- σ_P : Propositional terms.
- σ_I : Individual terms, including agents ($\sigma_A \subseteq \sigma_I$).
- $\tau_1 \rightarrow \tau_2$: Function types over these sorts.

The alphabet of $Log_A B$ includes:

- **Signature** Ω : A countable set of constant and function symbols with assigned types. Constants are 0-adic functions.
- **Variables** Ξ : Countably infinite, with $p_i \in \sigma_P$, $x_i \in \sigma_I$, $a_i \in \sigma_A$.
- **Syncategorematic symbols** Σ : Includes parentheses, commas, and $\forall$.

- **Logical symbols Λ :**

- Negation: $\neg \in \sigma_P \rightarrow \sigma_P$
- Conjunction, Disjunction: $\wedge, \vee \in \sigma_P \rightarrow \sigma_P \rightarrow \sigma_P$
- Belief: $B \in \sigma_A \rightarrow \sigma_P \rightarrow \sigma_P$

Semantics:

A $Log_A B$ structure is a tuple $\mathfrak{S} = \langle \mathcal{D}, \mathfrak{A}, b \rangle$, where [9]:

- $\mathcal{D}$ is a domain partitioned into:
 - P : A Boolean algebra of propositions.
 - A : A set of agents.
- $\mathfrak{A} = \langle P, +, \cdot, -, \perp, \top \rangle$ is a complete Boolean algebra [6].
- $b : A \times P \rightarrow P$ is the belief function.

3.4 Algebraic Logic of States

The Algebraic Logic of States ($Log_A S$) extends the algebraic treatment of belief ($Log_A B$) to model states as structured entities that **hold** over time rather than **occur** as events. Unlike modal logics, which represent states through accessibility relations, $Log_A S$ employs a Boolean algebraic framework where states are terms, not sentences [10].

3.4.1 Algebraic Representation of States

$Log_A S$ treats states as elements of a **Boolean algebra**, ensuring a well-defined structure for logical operations. Unlike traditional logics that represent states as sentences with truth conditions, $Log_A S$ defines states as algebraic terms, avoiding the syntactic distinction between formulas and terms [10].

Since $Log_A S$ models states as structured entities, it does not define truth in the classical sense. Instead, states are evaluated based on the **holding function** $h(s, t)$, which determines whether a state s holds at a time t . This eliminates the need for separate syntactic categories for propositions and removes redundancy in reasoning about state persistence and transformations [10].

Key Properties of $Log_A S$:

- **Boolean Algebra of States:** The set of states forms a Boolean algebra with operations for conjunction ($\wedge$), disjunction ($\vee$), and negation ($\neg$) [6].

- **State Persistence and Transition:** The persistence of a state over time is formally defined using the function $h(s, t)$, which determines whether a state s holds at a time t .
- **Separation from Events:** Unlike event-based models, Log_AS treats states as independent algebraic entities, rather than occurrences tied to transitions.

3.4.2 Syntax and Semantics of Log_AS

Syntax:

Log_AS is a many-sorted algebraic logic over terms with the base sorts [10]:

- σ_S : States
- σ_T : Time points
- σ_I : Individuals
- $\tau_1 \rightarrow \tau_2$: Function types over these sorts

Its alphabet includes:

- **Signature Ω :** Constant and function symbols with types from σ .
- **Variables Ξ :** Countably many $s_i \in \sigma_S$, $t_i \in \sigma_T$, $x_i \in \sigma_I$.
- **Symbols:**
 - Logical: $\neg, \wedge, \vee \in \sigma_S \rightarrow \dots \rightarrow \sigma_S$
 - Temporal: $\text{HoldsAt} \in \sigma_S \rightarrow \sigma_T \rightarrow \sigma_P$, $\prec \in \sigma_T \rightarrow \sigma_T \rightarrow \sigma_P$

Semantics:

A Log_AS model is $\mathfrak{S} = \langle \mathcal{D}, \mathfrak{A}, h, < \rangle$, where:

- $\mathcal{D}$ is sorted into states S and time points T .
- $\mathfrak{A} = \langle S, +, \cdot, -, \perp, \top \rangle$ is a Boolean algebra of states.
- $h : S \times T \rightarrow P$ defines when a state holds at time t , preserving Boolean structure:

$$h(-s, t) = -h(s, t), \quad h(s_1 \wedge s_2, t) = h(s_1, t) \wedge h(s_2, t)$$

- $< \subseteq T \times T$ is a strict linear order satisfying asymmetry, transitivity, irreflexivity, and seriality.

3.4.3 Temporal Stability of States

Log_AS classifies states based on **temporal stability**, partitioning them into distinct categories according to their persistence properties [10]. The classification is as follows:

- **Eternal States (ETER)**: States that hold at all times or at no time, with a fixed truth value across time.
- **Atemporal States (ATEMP)**: Eternal states that remain unchanged under temporal evaluation.
- **Permanent States (PERM)**: States that may start to hold, and once they do, never stop holding.
- **Co-Permanent States (CO-PERM)**: State of an event's occurrence being in the future.
- **Temporary States (TEMP)**: States that are not eternal, permanent, or co-permanent, possibly changing over time.
- **Transient States (TRANS)**: Temporary states that definitely stop holding at some times and start again at others.

These classes follow specific **closure properties under Boolean operations**, ensuring a rigorous algebraic characterization of state persistence [10]:

- **ETER and ATEMP** are closed under negation, conjunction, and disjunction, forming a Boolean subalgebra.
- **PERM and CO-PERM** are closed under finite conjunctions and disjunctions but not under negation.
- **TEMP and TRANS** are not closed under conjunction or disjunction, reflecting their transient nature.

This algebraic structure provides a mathematically rigorous foundation for reasoning about state persistence and transitions, improving upon classical modal and first-order logics [10].

3.4.4 Unifying States and Propositions

A central feature of Log_AS is its unification of **states** and **propositions** by identifying the class of **atemporal states (ATEMP)** with classical propositional terms [10]. This removes the need for a separate syntactic category, enabling uniform logical operations over both static and evolving truths.

3.4.5 Truth in $Log_A S$

In contrast to classical logic, truth in $Log_A S$ is defined algebraically via state holding. A state-term ϕ is **true** in a model $M_2 = \langle V, W_2 \rangle$ iff its interpretation belongs to a countably complete **ultrafilter** W_2 over the Boolean algebra of atemporal states:

$$\text{Tr}_{M_2}(\phi) \text{ iff } [[\phi]]^V \in W_2$$

An ultrafilter W_2 represents a bivalent world structure [2]: it contains $\top$, excludes $\perp$, is closed under conjunction, and for each state s , includes either s or its negation, but not both [10].

Truth Conditions

Let ϕ, ψ be state-denoting terms and $M_2 = \langle V, W_2 \rangle$. Then:

- $\text{Tr}_{M_2}(\neg\phi)$ iff $\text{Fl}_{M_2}(\phi)$.
- $\text{Tr}_{M_2}(\phi \wedge \psi)$ iff both $\text{Tr}_{M_2}(\phi)$ and $\text{Tr}_{M_2}(\psi)$.
- $\text{Tr}_{M_2}(\phi \vee \psi)$ iff either $\text{Tr}_{M_2}(\phi)$ or $\text{Tr}_{M_2}(\psi)$.
- $\text{Tr}_{M_2}(\forall x\phi)$ iff $\text{Tr}_{M_2^b}(\phi)$ for all $b \in D_{\sigma(x)}$, where M_2^b uses $V[b/x]$.

This ultrafilter-based semantics replaces modal satisfaction with algebraic state membership, ensuring logical consistency and compositional evaluation.

3.5 Conclusion

This chapter established the algebraic foundations necessary for modeling secrecy beyond the limitations of modal logic. By introducing the algebraic logic of belief ($Log_A B$) and the algebraic logic of states ($Log_A S$), we demonstrated how Boolean structures can replace relational semantics with term-based reasoning that is both expressive and compositional. $Log_A B$ provides a framework for modeling agent belief while avoiding paradoxes of self-reference, and $Log_A S$ extends this algebraic structure to model temporal persistence using state terms that hold over time.

Unlike modal logics, which rely on possible worlds and accessibility relations, the algebraic approach treats beliefs and states as algebraic functions, enabling precise evaluation of their behavior within a shared structure. This compositionality allows for clear interpretations of belief properties such as introspection, consistency, and distribution, and supports a principled classification of temporal states.

Together, these systems provide the core components required for constructing the Algebraic Logic of Secrets. The next chapter will synthesize these foundations into a unified logic capable of representing secrecy in multi-agent settings, capturing how secrets are held, revealed, and reasoned about over time.

Chapter 4

Algebraic Logic of Secrets

The Algebraic Logic of Secrets is a formal framework for reasoning about secrecy over time. It synthesizes components from multiple systems to address the structural representation of secrets, beliefs, intention, and revelation.

This logic builds on the formal modeling of secrecy proposed in [11], where secrets are described using a set of agents (secret keepers and nescients), a secretum (the proposition to be kept secret), a condition, and modal operators for belief, intention, and revelation [9, 10, 1, 11]. The paper defines secrecy in terms of what agents believe at a given time, and how that belief can be influenced by revelation or group membership.

To model time and change, we adopt the structure established in [1], where time is structured as a set of histories over a shared domain of time points. The VEL framework includes the key semantic constraints of strict past confluence, enabling reasoning about settled truths and alternative futures.

From [10], we incorporate the notion of *states* as algebraic entities. States are formal terms in a Boolean algebra that hold or fail to hold at specific time points [10, 6]. The classification of states into eternal, permanent, temporary, and atemporal enables structured reasoning about persistence and change. Importantly, atemporal states correspond to classical propositions, allowing us to define propositional reasoning as a subset of the broader algebraic state space.

Belief is modeled algebraically following [9], where belief is treated as a function satisfying KD45-like properties (monotonicity, consistency, introspection). This avoids reliance on possible-world semantics and enables a compositional and non-paradoxical representation of agent attitudes.

The resulting framework combines the algebraic treatment of belief and state evaluation with the temporal branching structure of VEL and the secrecy conditions defined in the Theory of Secrets.

4.1 The Language of $Log_A Sec$

Having established the algebraic foundations for belief and state, we now present the full formal language of the *Algebraic Logic of Secrets* ($Log_A Sec$). This language extends the algebraic term structure of $Log_A B$ and $Log_A S$ by incorporating additional syntactic and semantic components necessary for modeling secrecy. These include constructs for belief, intention, revelation, temporal structure, and group membership, all formalized within a unified many-sorted algebraic framework. The following subsections define the syntax, semantics, and interpretation of $Log_A Sec$ in detail.

4.1.1 Syntax and Semantics of $Log_A Sec$

Syntax:

The syntax of $Log_A Sec$ extends the many-sorted algebraic framework of $Log_A B$ and $Log_A S$, introducing additional sorts for groups and states while preserving the compositional term structure.

- σ_P : Terms that denote **propositions** (atemporal states).
- $\sigma_A \subseteq \sigma_I$: Terms that denote **individuals**, including a distinguished subset for **agents**.
- σ_G : Terms that denote **groups**.
- σ_T : Terms that denote **time points**.
- σ_S : Terms that denote **states**.

A $Log_A Sec$ alphabet is a union of four disjoint sets: the set of **constant and function symbols**, the set of **variables**, the set of **logical symbols**, and the set of **sorts**.

- **The signature Ω** : A non-empty, countable set of constant and function symbols, each assigned a syntactic type. It includes, among others, symbols such as:
 - $Mem \in \sigma_A \rightarrow \sigma_G \rightarrow \sigma_P$ (group membership)
 - $[] \in \sigma_A \rightarrow \sigma_G$ (singleton group formation)
 - $\sqcup, \sqcap \in \sigma_G \rightarrow \sigma_G \rightarrow \sigma_G$ (group union and intersection)
- **The variable set Ξ** : A countably infinite set $\{p_i, x_i, a_i, g_i, t_i, s_i\}_{i \in \mathbb{N}}$, where:
 - $p_i \in \sigma_P$ (denoting propositions),
 - $x_i \in \sigma_I$ (denoting individuals),
 - $a_i \in \sigma_A$ (denoting agents),
 - $g_i \in \sigma_G$ (denoting groups),
 - $t_i \in \sigma_T$ (denoting time points),
 - $s_i \in \sigma_S$ (denoting states),

- **The set of syncategorematic symbols Σ :** Includes parentheses, commas, and the universal quantifier symbols $\forall$, and $\exists$.
- **The set of logical symbols Λ :** Defined as the union of:
 1. $\{\neg\} \subseteq \sigma_P \rightarrow \sigma_P$ (Negation operator)
 2. $\{\wedge, \vee\} \subseteq \sigma_P \rightarrow \sigma_P \rightarrow \sigma_P$ (Conjunction and Disjunction)
 3. $\{B\} \subseteq \sigma_A \rightarrow \sigma_P \rightarrow \sigma_P$ (Belief function)
 4. $\{I\} \subseteq \sigma_A \rightarrow \sigma_P \rightarrow \sigma_P$ (Intention)
 5. $\{R\} \subseteq \sigma_A \rightarrow \sigma_P \rightarrow \sigma_P$ (Revelation)
 6. $\{\text{HoldsAt}\} \subseteq \sigma_P \rightarrow \sigma_T \rightarrow \sigma_P$ (Temporal holding function)
 7. $\{\boxtimes\} \subseteq \sigma_P \rightarrow \sigma_T \rightarrow \sigma_P$ (Strict past confluence)
 8. $\{\Box, \Diamond\} \subseteq \sigma_P \rightarrow \sigma_P$ (Necessity, possibility)
 9. $\{\prec\} \subseteq \sigma_T \rightarrow \sigma_T \rightarrow \sigma_P$ (Temporal ordering predicate)

A $Log_A Sec$ language with signature Ω is denoted by L_Ω . It is the smallest set of terms formed according to the following rules, where τ and τ_i ($i \in \mathbb{N}$) are terms in L_Ω .

- $\Xi \subset L_\Omega$: Every variable belongs to L_Ω .
- $c \in L_\Omega$, where $c \in \Omega$ is a constant symbol.
- $f(\tau_1, \dots, \tau_n) \in L_\Omega$, where $f \in \Omega$ is of type $\tau_1 \rightarrow \dots \rightarrow \tau_n \rightarrow \tau$ ($n > 0$) and τ_i is of type τ_i .
- $\neg\tau \in L_\Omega$, where $\tau \in \sigma_S$.
- $\tau_1 \wedge \tau_2 \in L_\Omega$, where $\tau_1, \tau_2 \in \sigma_P$ or $\tau_1, \tau_2 \in \sigma_S$.
- $\tau_1 \vee \tau_2 \in L_\Omega$, where $\tau_1, \tau_2 \in \sigma_P$ or $\tau_1, \tau_2 \in \sigma_S$.
- $\forall x(\tau) \in L_\Omega$, where $x \in \Xi$ and $\tau \in \sigma_P$ or $\tau \in \sigma_S$.
- $B(\tau_1, \tau_2) \in L_\Omega$, where $\tau_1 \in \sigma_A$, $\tau_2 \in \sigma_P$.
- $I(\tau_1, \tau_2) \in L_\Omega$, where $\tau_1 \in \sigma_A$, $\tau_2 \in \sigma_P$.
- $R(\tau_1, \tau_2) \in L_\Omega$, where $\tau_1 \in \sigma_A$, $\tau_2 \in \sigma_P$.
- $\text{HoldsAt}(\tau_1, \tau_2) \in L_\Omega$, where $\tau_1 \in \sigma_P$ and $\tau_2 \in \sigma_T$.
- $(\tau_1 \prec \tau_2) \in L_\Omega$, where $\tau_1, \tau_2 \in \sigma_T$.
- $\boxtimes\tau \in L_\Omega$, where $\tau \in \sigma_P$.
- $\Box\tau, \Diamond\tau \in L_\Omega$, where $\tau \in \sigma_P$.

- $Mem(\tau_1, \tau_2) \in L_\Omega$, where $\tau_1 \in \sigma_A$ and $\tau_2 \in \sigma_G$.
- $[\tau_1] \in L_\Omega$, where $\tau_1 \in \sigma_A$.
- $\tau_1 \sqcup \tau_2 \in L_\Omega$, where $\tau_1, \tau_2 \in \sigma_G$.
- $\tau_1 \sqcap \tau_2 \in L_\Omega$, where $\tau_1, \tau_2 \in \sigma_G$.

As usual, terms involving $\Rightarrow$, $\Leftrightarrow$, and $\exists$ may be introduced as abbreviations in the standard way.

Semantics:

A $Log_A Sec$ structure is a tuple $\mathfrak{S} = \langle \mathcal{D}, \mathfrak{A}, b, i, r, h, < \rangle$, where:

- $\mathcal{D}$ is a domain partitioned into:
 - P : A Boolean algebra of propositions (atemporal states).
 - A : A set of agents.
 - G : A set of groups.
 - H : A set of histories.
 - T : A set of time points.
 - S : A set of states.
- $\mathfrak{A} = \langle S, +, \cdot, -, \perp, \top \rangle$ is a complete Boolean algebra [6].
- $b : A \times P \times H \times T \rightarrow P$ is the belief function.
- $i : A \times P \times H \times T \rightarrow P$ is the intention function.
- $r : A \times P \times H \times T \rightarrow P$ is the revelation function.
- $h : P \times T \times H \rightarrow P$ is the holding function.
- $< : T \times T \rightarrow P$ is the linear ordering function.

Valuation and Interpretation

A valuation $\mathcal{V}$ in L_Ω assigns elements of the appropriate domain to variables and constants, and interprets function symbols by sorted functions.

- $\mathcal{V}(x, h, t) \in \mathcal{D}_\sigma$ if $x \in \Xi$ is a variable of sort σ .
- $\mathcal{V}(c, h, t) \in \mathcal{D}_\sigma$ if $c \in \Omega$ is a constant of sort σ .
- $\mathcal{V}(f)$ is a function $\mathcal{D}_{\tau_1} \times \cdots \times \mathcal{D}_{\tau_n} \rightarrow \mathcal{D}_\tau$, if $f \in \Omega$ is a function symbol of type $\tau_1 \rightarrow \cdots \rightarrow \tau_n \rightarrow \tau$.

Let $t \in \sigma_T$ denote the current time point of evaluation. The interpretation function $\llbracket \cdot \rrbracket_{\mathcal{V}}^{h,t}$ defines the meaning of terms and formulas in the context of a history h and time t :

- $\llbracket x \rrbracket_{\mathcal{V}}^{h,t} = \mathcal{V}(x, h, t)$ for $x \in \Xi$.
- $\llbracket c \rrbracket_{\mathcal{V}}^{h,t} = \mathcal{V}(c, h, t)$ for $c \in \Omega$.
- $\llbracket f(\tau_1, \dots, \tau_n) \rrbracket_{\mathcal{V}}^{h,t} = \mathcal{V}(f)(\llbracket \tau_1 \rrbracket_{\mathcal{V}}^{h,t}, \dots, \llbracket \tau_n \rrbracket_{\mathcal{V}}^{h,t})$ for $f \in \Omega$.
- $\llbracket \neg \tau \rrbracket_{\mathcal{V}}^{h,t} = -\llbracket \tau \rrbracket_{\mathcal{V}}^{h,t}$, for $\tau \in \sigma_P$ or $\tau \in \sigma_S$.
- $\llbracket \tau_1 \wedge \tau_2 \rrbracket_{\mathcal{V}}^{h,t} = \llbracket \tau_1 \rrbracket_{\mathcal{V}}^{h,t} \cdot \llbracket \tau_2 \rrbracket_{\mathcal{V}}^{h,t}$, for $\tau_1, \tau_2 \in \sigma_P$.
- $\llbracket \tau_1 \vee \tau_2 \rrbracket_{\mathcal{V}}^{h,t} = \llbracket \tau_1 \rrbracket_{\mathcal{V}}^{h,t} + \llbracket \tau_2 \rrbracket_{\mathcal{V}}^{h,t}$, for $\tau_1, \tau_2 \in \sigma_P$.
- $\llbracket \forall x(\tau) \rrbracket_{\mathcal{V}}^{h,t} = \prod_{a \in \mathcal{D}_{\sigma(x)}} \llbracket \tau \rrbracket_{\mathcal{V}[a/x]}^{h,t}$, where x is of sort σ .
- $\llbracket B(\tau_1, \tau_2) \rrbracket_{\mathcal{V}}^{h,t} = b(\llbracket \tau_1 \rrbracket_{\mathcal{V}}^{h,t}, \llbracket \tau_2 \rrbracket_{\mathcal{V}}^{h,t})$, for $\tau_1 \in \sigma_A, \tau_2 \in \sigma_P$.
- $\llbracket I(\tau_1, \tau_2) \rrbracket_{\mathcal{V}}^{h,t} = i(\llbracket \tau_1 \rrbracket_{\mathcal{V}}^{h,t}, \llbracket \tau_2 \rrbracket_{\mathcal{V}}^{h,t})$, for $\tau_1 \in \sigma_A, \tau_2 \in \sigma_P$.
- $\llbracket R(\tau_1, \tau_2) \rrbracket_{\mathcal{V}}^{h,t} = r(\llbracket \tau_1 \rrbracket_{\mathcal{V}}^{h,t}, \llbracket \tau_2 \rrbracket_{\mathcal{V}}^{h,t})$, for $\tau_1 \in \sigma_A, \tau_2 \in \sigma_P$.
- $\llbracket HoldsAt(\tau_1, \tau_2) \rrbracket_{\mathcal{V}}^{h,t} = h(\llbracket \tau_1 \rrbracket_{\mathcal{V}}^{h,t}, \llbracket \tau_2 \rrbracket_{\mathcal{V}}^{h,t})$, for $\tau_1 \in \sigma_P, \tau_2 \in \sigma_T$.
- $\llbracket \tau_1 \prec \tau_2 \rrbracket_{\mathcal{V}}^{h,t} = (\llbracket \tau_1 \rrbracket_{\mathcal{V}}^{h,t} < \llbracket \tau_2 \rrbracket_{\mathcal{V}}^{h,t})$, for $\tau_1, \tau_2 \in \sigma_T$.
- $\llbracket \boxtimes \phi \rrbracket_{\mathcal{V}}^{h,t}$ is true iff ϕ holds in all strict predecessors of $\langle h, t \rangle$, for $\phi \in \sigma_P$.
- $\llbracket \Box \phi \rrbracket_{\mathcal{V}}^{h,t}$ holds at all accessible points from $\langle h, t \rangle$; $\llbracket \Diamond \phi \rrbracket_{\mathcal{V}}^{h,t}$ holds at some accessible point, for $\phi \in \sigma_P$.
- $\llbracket Mem(\tau_1, \tau_2) \rrbracket_{\mathcal{V}}^{h,t} = \mathcal{V}(Mem)(\llbracket \tau_1 \rrbracket_{\mathcal{V}}^{h,t}, \llbracket \tau_2 \rrbracket_{\mathcal{V}}^{h,t})$, for $\tau_1 \in \sigma_A, \tau_2 \in \sigma_G$.
- $\llbracket [\tau_1] \rrbracket_{\mathcal{V}}^{h,t} = \mathcal{V}([\])(\llbracket \tau_1 \rrbracket_{\mathcal{V}}^{h,t})$, for $\tau_1 \in \sigma_A$.
- $\llbracket \tau_1 \sqcup \tau_2 \rrbracket_{\mathcal{V}}^{h,t} = \mathcal{V}(\sqcup)(\llbracket \tau_1 \rrbracket_{\mathcal{V}}^{h,t}, \llbracket \tau_2 \rrbracket_{\mathcal{V}}^{h,t})$, for $\tau_1, \tau_2 \in \sigma_G$.
- $\llbracket \tau_1 \sqcap \tau_2 \rrbracket_{\mathcal{V}}^{h,t} = \mathcal{V}(\sqcap)(\llbracket \tau_1 \rrbracket_{\mathcal{V}}^{h,t}, \llbracket \tau_2 \rrbracket_{\mathcal{V}}^{h,t})$, for $\tau_1, \tau_2 \in \sigma_G$.

4.1.2 $Log_A Sec$ Axioms

The axiom system of $Log_A Sec$ encodes belief, intention, revelation, and group membership as algebraic functions, providing a compositional alternative to the modal axiom system Σ defined in [11]. Belief (B) satisfies the standard KD45 axioms, reflecting consistency, positive introspection, and negative introspection, whereas intention (I) satisfies KD axioms, ensuring consistency without the stronger introspection conditions. Each axiom is formulated over Boolean terms within the state algebra and is evaluated relative to histories and time points.

Belief Axioms (KD45)

- (B1) $B(a, p) \wedge B(a, p \rightarrow q) \rightarrow B(a, q)$
- (B2) $B(a, p) \rightarrow \neg B(a, \neg p)$
- (B3) $B(a, p) \rightarrow B(a, B(a, p))$
- (B4) $\neg B(a, p) \rightarrow B(a, \neg B(a, p))$
- (B5) If $\vdash p$ then $B(a, p)$

Intention Axioms (KD)

- (I1) $I(a, p) \wedge I(a, p \rightarrow q) \rightarrow I(a, q)$
- (I2) $I(a, p) \rightarrow \neg I(a, \neg p)$
- (I3) If $\vdash p$ then $I(a, p)$

Belief–Intention Bridge Axioms These axioms govern the relation between belief and intention, ensuring agents are introspectively aware of their own commitments:

- (IB1) $\neg I(a, p) \rightarrow B(a, \neg I(a, p))$
- (IB2) $B(a, \neg I(a, p)) \rightarrow \neg I(a, p)$
- (IB3) $I(a, p) \rightarrow B(a, I(a, p))$
- (IB4) $B(a, I(a, p)) \rightarrow I(a, p)$
- (IB5) $I(a, p) \rightarrow \neg B(a, \neg p)$

Revelation Axioms These axioms define how evidence-based revelation behaves, including consistency and closure under implication:

- (R1) If $\vdash p$ then $R(a, p)$
- (R2) If $\vdash p$ then $\neg R(a, \neg p)$
- (R3) If $\vdash p \rightarrow q$ then $R(a, p) \rightarrow R(a, q)$
- (R4) $R(a, \exists b(R(b, p))) \rightarrow R(a, p)$

Belief–Revelation Bridge Axioms These axioms connect the modalities of belief and revelation:

$$\begin{aligned} \text{(BR1)} \quad & [B(a, p \rightarrow q) \wedge \neg B(a, \neg p)] \rightarrow [R(a, p) \rightarrow R(a, q)] \\ \text{(BR2)} \quad & R(a, p) \rightarrow B(a, R(a, p)) \end{aligned}$$

Group Membership Axioms Group-related properties follow the standard algebraic definitions of singleton, union, and intersection:

$$\begin{aligned} \text{(G1)} \quad & Mem(a, [b]) \leftrightarrow (a = b) \\ \text{(G2)} \quad & Mem(a, G_1 \sqcup G_2) \leftrightarrow Mem(a, G_1) \vee Mem(a, G_2) \\ \text{(G3)} \quad & Mem(a, G_1 \sqcap G_2) \leftrightarrow Mem(a, G_1) \wedge Mem(a, G_2) \end{aligned}$$

These axioms form the core of Log_ASec 's reasoning capabilities, ensuring consistency, introspective properties for belief and intention, evidence-based propagation through revelation, and dynamic group semantics necessary for secrecy modeling over time and histories.

4.1.3 Log_ASec Theorems

Based on the axiom system defined above, this section presents a set of theorems that hold in Log_ASec . These theorems reflect structural properties of secrecy, belief propagation, revelation, and group interactions. Many are inspired by or correspond to results in [11], though they are here reformulated in our algebraic framework.

$$\begin{aligned} \text{(T1)} \quad & R(a, p \wedge q) \rightarrow R(a, p) \wedge R(a, q) \\ \text{(T2)} \quad & B(a, p) \rightarrow R(a, p) \\ \text{(T3)} \quad & R(a, p) \leftrightarrow R(a, R(a, p)) \\ \text{(T4)} \quad & B(a, p) \rightarrow B(a, R(a, p)) \\ \text{(T5)} \quad & B(a, R(a, p)) \rightarrow R(a, p) \\ \text{(T6)} \quad & B(a, \neg R(a, p)) \rightarrow \neg R(a, p) \\ \text{(T7)} \quad & R(a, B(a, p)) \rightarrow B(a, R(a, p)) \end{aligned}$$

These theorems demonstrate how belief and revelation interact in Log_ASec . For example, (T2) indicates that belief entails revelation, whereas (T3)–(T5) show introspective and closure properties of revelation.

4.1.4 Selected Theorem Proofs

This section presents formal proofs of Theorems (T1)–(T7) [11], which characterize key relationships between belief, revelation, and their interactions in the *Log_ASec* framework. Each proof follows directly from the axiom system introduced earlier, with references to specific axioms and previously established theorems as needed. These results illustrate the logical properties that hold in the presence of full logical omniscience (the K axiom).

Theorem (T1) $R(a, p \wedge q) \rightarrow R(a, p) \wedge R(a, q)$

Proof.

1. $R(a, p \wedge q)$ [assumption]
2. $p \wedge q \rightarrow p$ [tautology]
3. $R(a, p)$ [2, R3]
4. $p \wedge q \rightarrow q$ [tautology]
5. $R(a, q)$ [4, R3]
6. $R(a, p) \wedge R(a, q)$ [3,5]
7. $R(a, p \wedge q) \rightarrow R(a, p) \wedge R(a, q)$ [1,6]

Theorem (T2) $B(a, p) \rightarrow R(a, p)$

Proof.

1. $B(a, p)$ [assumption]
2. $B(a, \top \rightarrow p)$ [tautology]
3. $B(a, \top)$ [B5]
4. $\neg B(a, \neg \top)$ [3, B2]
5. $B(a, \top \rightarrow p) \wedge \neg B(a, \neg \top)$ [2,4]
6. $R(a, \top) \rightarrow R(a, p)$ [5, BR1]
7. $R(a, \top)$ [R1]
8. $R(a, p)$ [6,7]
9. $B(a, p) \rightarrow R(a, p)$ [1,8]

Theorem (T3) $R(a, p) \leftrightarrow R(a, R(a, p))$

Proof.

1. $R(a, p)$ [assumption]
2. $R(a, p) \rightarrow B(a, R(a, p))$ [BR2]
3. $B(a, R(a, p))$ [1,2]
4. $R(a, R(a, p))$ [3, Theorem T2]
5. $R(a, p) \rightarrow R(a, R(a, p))$ [1,4]
6. $R(a, R(a, p))$ [assumption]
7. $R(a, p)$ [6, R4]
8. $R(a, R(a, p)) \rightarrow R(a, p)$ [6,7]

Therefore, $R(a, p) \leftrightarrow R(a, R(a, p))$.

Theorem (T4) $B(a, p) \rightarrow B(a, R(a, p))$

Proof.

1. $B(a, p)$ [assumption]
2. $B(a, B(a, p))$ [1, B3]
3. $B(a, B(a, p) \rightarrow R(a, p))$ [B5, Theorem T2]
4. $B(a, R(a, p))$ [2,3, B1]
5. $B(a, p) \rightarrow B(a, R(a, p))$ [1,4]

Theorem (T5) $B(a, R(a, p)) \rightarrow R(a, p)$

Proof.

1. $B(a, R(a, p))$ [assumption]
2. $R(a, R(a, p))$ [1, Theorem T2]
3. $R(a, p)$ [2, Theorem T3]
4. $B(a, R(a, p)) \rightarrow R(a, p)$ [1,3]

Theorem (T6) $B(a, \neg R(a, p)) \rightarrow \neg R(a, p)$

Proof.

1. $B(a, \neg R(a, p))$ [assumption]
2. $\neg B(a, R(a, p))$ [1, B2]
3. $\neg R(a, p)$ [2, BR2]
4. $B(a, \neg R(a, p)) \rightarrow \neg R(a, p)$ [1,3]

Theorem (T7) $R(a, B(a, p)) \rightarrow B(a, R(a, p))$

Proof.

1. $R(a, B(a, p))$ [assumption]
2. $B(a, p) \rightarrow R(a, p)$ [Theorem T2]
3. $R(a, R(a, p))$ [1,2, R3]
4. $R(a, p)$ [3, Theorem T3]
5. $B(a, R(a, p))$ [4, BR2]
6. $R(a, B(a, p)) \rightarrow B(a, R(a, p))$ [1,5]

4.1.5 Implications of Removing Logical Omniscience

In classical modal logic, the K axiom enforces logical omniscience: the assumption that agents believe all logical consequences of their beliefs. While useful for formal derivations, this assumption is widely regarded as unrealistic in practical multi-agent systems, where agents are computationally bounded and do not possess perfect deductive capabilities. Real-world agents typically lack the capacity for unbounded reasoning or tracking all logical consequences of their beliefs [8].

To address this, we remove the following axiom from $Log_A Sec$:

- **B1:** $B(a, p) \wedge B(a, p \rightarrow q) \rightarrow B(a, q)$

This is the K axiom. It enables deductive closure of belief by allowing agents to infer $B(a, q)$ whenever they believe p and $p \rightarrow q$. By removing it, we model agents whose reasoning is intentionally limited, aligning the logic with more realistic assumptions about cognition and information processing.

The removal of B1 affects only Theorem (T4): $B(a, p) \rightarrow B(a, R(a, p))$. The proof fails at step 4, where B1 is used to derive $B(a, R(a, p))$ from nested beliefs. All other theorems in this section remain provable under the revised axiom system.

4.2 Conclusion

In this chapter, we presented the Algebraic Logic of Secrets (*Log_ASec*), a formal system designed to model secrecy across time, agents, and evolving group structures. By integrating the algebraic modeling of belief and state from prior logics with the temporal reasoning of Versatile Event Logic and the secrecy conditions defined in the Commonsense Theory of Secrets, *Log_ASec* offers a compositional and expressive framework for reasoning about how secrets are maintained, revealed, and believed.

We introduced the syntax, semantics, and axiomatic foundations of *Log_ASec*, capturing belief, intention, revelation, and group membership using sorted algebraic functions. The resulting theorems demonstrated key relationships between these concepts, such as how belief supports revelation and how agents reason about their own knowledge and commitments.

We also considered a crucial limitation of classical belief models: logical omniscience. By removing the *K* axiom from the belief system, we explored a more realistic variant of *Log_ASec* where agents are not automatically logically closed. This modification led to the loss of several belief-revelation theorems, highlighting the tradeoff between idealized reasoning and practical agent modeling.

Overall, *Log_ASec* provides a flexible foundation for analyzing secrecy in multi-agent systems. Its structure allows for further refinements, including the incorporation of bounded rationality, dynamic group updates, or probabilistic extensions, making it a useful tool for both theoretical inquiry and real-world applications in secure information systems and explainable AI.

Chapter 5

Conclusion & Future Work

This final chapter summarizes the main contributions of the thesis and reflects on their implications for both theory and practice. It also outlines possible directions for future work and extensions of the presented logic.

5.1 Conclusion

The Algebraic Logic of Secrets (*Log_ASec*) introduced in this thesis provides a formal framework for modeling secrecy in multi-agent systems. By synthesizing ideas from modal logic, algebraic representations of belief and state, and the Commonsense Theory of Secrets, we developed a logic that captures the structural, temporal, and epistemic dimensions of secrecy.

Unlike classical modal approaches, *Log_ASec* avoids possible-world semantics and instead adopts a compositional, term-based algebraic logic. This shift enables a finer-grained representation of reasoning processes, particularly in contexts that demand transparency or bounded rationality. The logic accounts for key elements of secrecy, including belief, intention, revelation, group membership, and time, and expresses them within a unified and extensible language grounded in many-sorted algebraic structures.

Theoretically, *Log_ASec* contributes a novel unification of existing systems: it merges the algebraic logic of belief (*Log_AB*), the algebraic logic of states (*Log_AS*), and Versatile Event Logic (VEL) within a single formalism tailored to secrecy. It extends these frameworks by introducing multi-sorted types, temporal branching, and new operators such as revelation and membership, all governed by algebraic rather than relational semantics. The axiom system demonstrates how belief and intention interact with revelation and secrecy, and the theorems formalize epistemic dependencies previously handled only via modal reasoning.

Furthermore, the framework explicitly models the effects of removing logical omniscience, thereby contributing to ongoing efforts in formal logic to capture resource-bounded reasoning. This allows *Log_ASec* to reflect more realistic agent behavior while preserving algebraic compositionality.

Overall, this work makes both a theoretical and practical contribution. It expands the space of formal logics by constructing a non-paradoxical, compositional, and temporally expressive logic of secrecy. Simultaneously, it provides a foundation for future implementations in explainable AI, secure multi-agent interaction, and formal cognitive architectures.

5.2 Future Work

Several directions remain open for future exploration. One natural extension is to explore and formally prove the remaining theorems presented in [11], which were not addressed in the *Log_ASec* framework. While this thesis proves a representative subset, a full proof catalog would provide a more comprehensive validation of the logic and may reveal deeper structural properties.

Another promising direction is the implementation of *Log_ASec* in a logic programming environment, such as Prolog. Given the term-based and rule-driven structure of algebraic logic, such an implementation could enable simulation of agent reasoning and allow experimentation with concrete secrecy scenarios. This could be especially useful for testing how agents manage secrets under different conditions, or for prototyping secrecy-aware components in multi-agent systems.

Additionally, further refinement of the logic could involve modeling resource-bounded reasoning by weakening belief axioms even further, or by introducing probabilistic extensions. More expressive group structures, dynamic membership, and richer representations of intention are also potential areas for development.

Overall, this framework offers a strong foundation for ongoing research into secrecy modeling, both from a formal and applied perspective.

Appendix

Appendix A

Propositional and Modal Logic

Logic provides a formal framework for reasoning about truth and inference. Among the most fundamental logical systems are **propositional logic** and **modal logic**. While propositional logic deals strictly with truth-functional relationships between propositions, modal logic extends this framework by introducing modalities such as necessity and possibility. This section explores the differences between these two logical systems, highlighting their expressive power, limitations, and applications.

A.1 Propositional Logic

Propositional logic, also known as sentential logic, is a formal system that evaluates logical statements based on their truth values. It consists of atomic propositions, logical connectives ($\wedge$ for AND, $\vee$ for OR, $\neg$ for NOT, and $\rightarrow$ for IMPLIES), and rules of inference. A proposition is either true or false, and its truth value is determined purely by the structure of the logical expression.

Propositional logic offers several strengths that make it a fundamental tool in formal reasoning. Its simplicity provides a straightforward method for reasoning about truth-functional statements. In addition, it has well-defined syntax and semantics, where logical operators and truth tables precisely determine how propositions combine. Another key advantage is its efficiency in computation, making it widely used in automated reasoning, satisfiability checking, and circuit design. However, propositional logic also has notable limitations. It lacks the ability to represent modal notions such as necessity and possibility, preventing the expression of statements like "it is necessarily true that" or "it is possible that" [5]. Furthermore, unlike predicate logic, it does not support quantification, restricting its capacity to reason about variables or objects. Another issue arises from material implication paradoxes, as classical propositional logic allows any statement to be inferred from a falsehood, leading to counterintuitive results, for example, "If the moon is made of cheese, then $2+2=5$ " or "If unicorns exist, then I am the president" both hold true under material implication [5, 3].

A.2 Modal Logic

Modal logic extends propositional logic by introducing **modal operators** that qualify the truth of a statement. The main modal operators are $\Box$ (necessity) and $\Diamond$ (possibility). A formula $\Box\phi$ means that ϕ is necessarily true, while $\Diamond\phi$ means that ϕ is possibly true. These operators satisfy the equivalence $\Box\phi \equiv \neg\Diamond\neg\phi$ [5].

The meaning of these operators is formally defined using **possible-worlds semantics**, a framework introduced by Saul Kripke [5, 2]. In this approach, the truth of a statement is evaluated not just in a single world, but across a set of related possible worlds [5]. A modal statement's truth depends on accessibility relations between worlds, which specify how worlds are connected. Different modal logics arise depending on constraints on these relations, such as reflexivity, transitivity, and symmetry.

Modal logic has broad applications in various fields, including philosophy, linguistics, artificial intelligence, and computer science. It is widely used in epistemic logic to model knowledge and belief, in temporal logic to reason about time, and in deontic logic to analyze obligations and permissions. The expressiveness of modal logic makes it a powerful tool for formalizing reasoning beyond classical propositional logic [5].

Modal logic offers several advantages that extend beyond the capabilities of propositional logic. Its expressiveness allows for reasoning about necessity, possibility, knowledge, belief, and obligation, making it a powerful framework for various logical analyses. This broad applicability is evident in fields such as epistemic logic, which deals with knowledge representation, deontic logic, which formalizes obligations, and temporal logic, which handles reasoning about time-based statements. Additionally, modal logic helps overcome certain paradoxes of material implication found in classical propositional logic by incorporating strict implication ($\Box(p \rightarrow q)$), thus avoiding some counterintuitive results [5]. However, these advantages come at the cost of increased complexity. The introduction of modal operators necessitates a more sophisticated semantics based on possible world models, making the formalism more intricate. Furthermore, while many problems in propositional logic can be solved efficiently, reasoning within modal logic often demands significantly greater computational resources.

Propositional logic and modal logic serve distinct but complementary roles in formal reasoning. Propositional logic offers a foundation for analyzing statements based on their truth values, making it practical for computational tasks. However, it lacks the ability to express modal notions such as necessity and possibility. Modal logic addresses this limitation by extending the framework with modal operators, allowing for reasoning about knowledge, time, and obligations. While modal logic is more expressive, it also introduces additional complexity in interpretation and computation. Understanding both logics provides a richer framework for applications in artificial intelligence, philosophy, and formal verification.

A.2.1 Formal Representation in Modal Logic

Propositional Modal Logic (PML)

Propositional Modal Logic (PML) extends classical propositional logic by introducing modal operators that allow reasoning about necessity and possibility. A well-formed formula (WFF) in PML consists of atomic propositions, such as p, q, r , combined using classical logical connectives $\neg, \wedge, \vee, \Rightarrow$. Additionally, PML includes the modal operators $\Box$ (necessity) and $\Diamond$ (possibility), which enable the expression of statements about what is necessarily or possibly true. These modal operators enrich the expressiveness of classical logic, making it useful for formalizing knowledge, obligations, and temporal reasoning [2].

Some axioms of PML:

- Distribution Axiom [K]: $\Box(\phi \Rightarrow \psi) \Rightarrow (\Box\phi \Rightarrow \Box\psi)$
- Reflexivity Axiom [T]: $\Box\phi \Rightarrow \phi$
- Seriality Axiom [D]: $\Box\phi \Rightarrow \Diamond\phi \ (\equiv \Box\phi \Rightarrow \neg\Box\neg\phi)$
- Transitivity Axiom [4]: $\Box\phi \Rightarrow \Box\Box\phi$
- Euclidean Axiom [5]: $\Diamond\phi \Rightarrow \Box\Diamond\phi$
- Symmetry Axiom [B]: $\phi \Rightarrow \Box\Diamond\phi$

While PML allows reasoning about necessity and possibility at a propositional level, it does not account for objects or individuals within these worlds. FOML extends this capability by introducing quantification, allowing for richer formalization of dynamic secrecy models. This extension is crucial in contexts where secrets involve multiple agents, evolving knowledge states, and temporal dynamics.

First-Order Modal Logic (FOML)

First-Order Modal Logic (FOML) extends modal logic by incorporating quantifiers such as $\forall$ (universal quantification) and $\exists$ (existential quantification), allowing reasoning about objects across possible worlds. Its semantics are based on a Kripke structure (W, R, M) , where $M = (U, I_F, I_P)$ represents a first-order structure. Here, U is a non-empty domain of discourse, I_F maps function symbols to interpretations across possible worlds, and I_P assigns predicates to relations over U in each world. A key challenge in FOML is handling quantification across worlds, as illustrated by the Barcan formula $\forall x\Box\phi \Rightarrow \Box\forall x\phi$, which has significant implications in modal semantics [5, 2]. Modal logic has applications in epistemic logic, deontic logic, and temporal logic [11]. The next chapters will examine its role in modeling secrets algebraically and explore its connections to algebraic logic and formal reasoning. The subsequent chapters will analyze its formal properties, algebraic structure, and applications to the study of secrets.

FOML enables the modeling of secrecy by allowing quantification over possible worlds, which is crucial in formalizing secret distribution and access. This allows us to reason about which agents know a secret, how knowledge propagates, and under what conditions secrets remain hidden or become revealed.

A.2.2 Interpretation of Modal Logic

Semantics of PML

The interpretation function $\llbracket \cdot \rrbracket$ assigns to each $\mathcal{L}_P$ WFF a truth value with respect to some model $\mathcal{M} = (\mathcal{W}, \mathcal{R}, \mathcal{A})$ and some world $w \in \mathcal{W}$.

- $\llbracket T \rrbracket^{\mathcal{M}, w} = \top$.
- $\llbracket F \rrbracket^{\mathcal{M}, w} = \perp$.
- $\llbracket P \rrbracket^{\mathcal{M}, w} = \mathcal{A}(P)(w)$, for every $P \in \mathcal{P}$.

The interpretation of modal formulas relies on a structured way to evaluate their truth across multiple possible worlds. This structure is provided by Kripke models, which define how worlds are related and how truth values propagate between them.

Kripke Model

A Kripke model consists of a frame $K = (W, R)$, where W is a set of possible worlds and R is an accessibility relation between them. The valuation function V assigns truth values to each statement in every world. The interpretation of formulas follows these rules [5]:

- For $\phi, \psi \in \mathcal{L}_{\mathcal{V}}$
 1. $\llbracket \neg \phi \rrbracket^{\mathcal{M}, w} = \top$ iff $\llbracket \phi \rrbracket^{\mathcal{M}, w} = \perp$.
 2. $\llbracket (\phi \wedge \psi) \rrbracket^{\mathcal{M}, w} = \top$ iff $\llbracket \phi \rrbracket^{\mathcal{M}, w} = \top$ and $\llbracket \psi \rrbracket^{\mathcal{M}, w} = \top$.
 3. $\llbracket (\phi \vee \psi) \rrbracket^{\mathcal{M}, w} = \top$ iff $\llbracket \phi \rrbracket^{\mathcal{M}, w} = \top$ or $\llbracket \psi \rrbracket^{\mathcal{M}, w} = \top$.
 4. $\llbracket (\phi \Rightarrow \psi) \rrbracket^{\mathcal{M}, w} = \top$ iff $\llbracket \phi \rrbracket^{\mathcal{M}, w} = \perp$ or $\llbracket \psi \rrbracket^{\mathcal{M}, w} = \top$.
 5. $\llbracket (\phi \Leftrightarrow \psi) \rrbracket^{\mathcal{M}, w} = \top$ iff $\llbracket \phi \rrbracket^{\mathcal{M}, w} = \llbracket \psi \rrbracket^{\mathcal{M}, w}$.
 6. $\llbracket \Box \phi \rrbracket^{\mathcal{M}, w} = \top$ iff for every $w' \in \mathcal{W}$ such that wRw' , $\llbracket \phi \rrbracket^{\mathcal{M}, w'} = \top$.
 7. $\llbracket \Diamond \phi \rrbracket^{\mathcal{M}, w} = \top$ iff for some $w' \in \mathcal{W}$ such that wRw' , $\llbracket \phi \rrbracket^{\mathcal{M}, w'} = \top$.

The properties of R determine different modal systems. For example, reflexivity (wRw for all $w \in W$) defines system T , while transitivity ($wRv \wedge vRu \Rightarrow wRu$) leads to system $S4(KT4)$ [5].

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